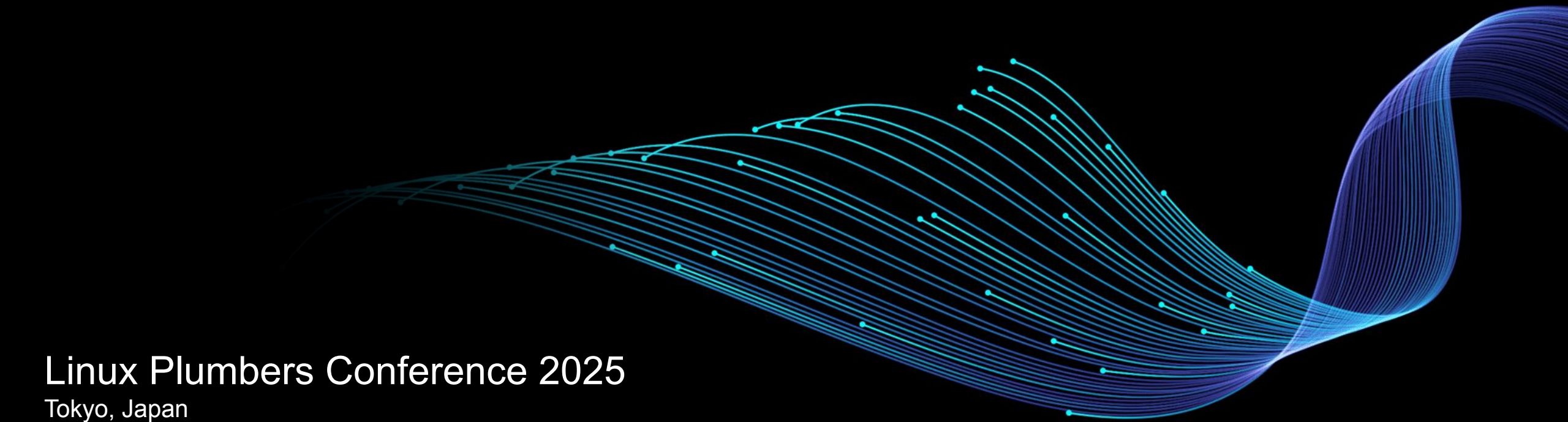
SAMSUNG

The singularity is here for testing: genai advancements on kdevops



Linux Plumbers Conference 2025

Tokyo, Japan

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Daniel Gomez <da.gomez@samsung.com>

Samsung Semiconductor

December 12, 2025

kdevops: intro

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  - Linux distribution agnostic

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    - **But docker containers are not suitable for kernel testing !**

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    - guestfs for local virtualization
    - terraform or opentofu
      - OCI
      - AWS
      - Azure
      - GCE
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      - GCE
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      - Neoclouds: Lambda Labs & DataCrunch (Verdi)
    - Bare metal
  - All parts of kdevops have been designed with:
    - Automation & Scaling philosophy

kdevops: at its core

- What is kdevops:
  - Kernel automation framework **for Linux kernel developers** **by kernel developers**



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    - blktests
    - selftests
    - pynfs
    - nfstest
    - ltp
    - gitr
    - sysbench



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    - fio-tests



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Plugin architecture: custom or even proprietary workflows

- knlp: Kernel development style Natural Language Processing R&D



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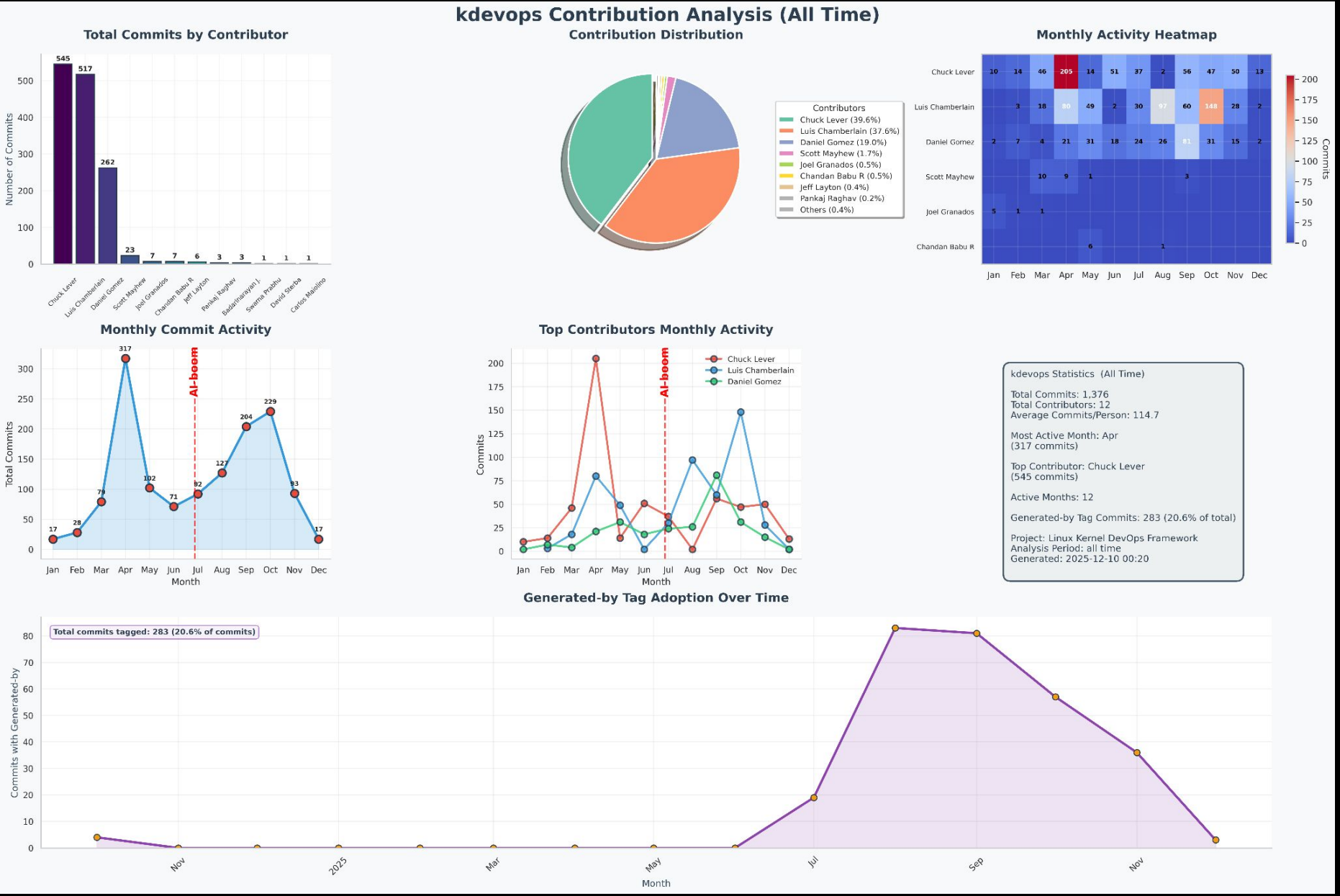
Plugin architecture: custom or even proprietary workflows

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rcloud: leverage terraform for your own servers

Simplified cloud solution for enhancing CI determinism
Implemented in Rust by Generative AI
Alternative to OpenStack, Ubicloud

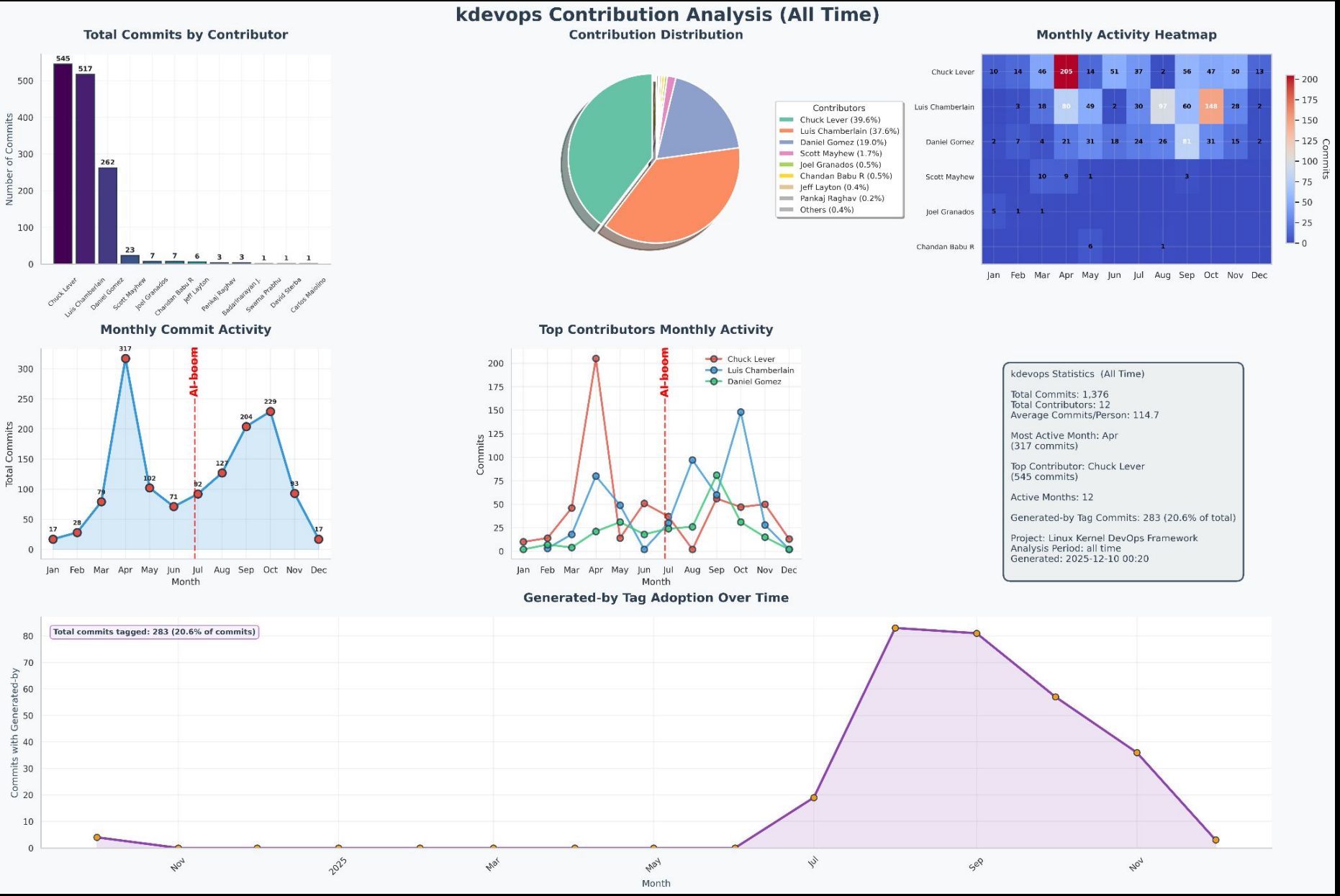


AI impact

is

transformative

Generated-by



AI impact

can be more

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AI impact

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How?

Generated-by

kdevops: how do we make filesystem tests easier?

SAMSUNG

kdevops: how do we make filesystem tests easier?

SAMSUNG

Kconfig

```
config SYSBENCH_EXT4_SECTION_4K_BIGALLOC_16K_CMD
        string
        output yaml
        depends on SYSBENCH_EXT4_SECTION_4K_BIGALLOC_16K
        default "mkfs.ext4 -F -b 4k -O bigalloc -C 16k"

config SYSBENCH_EXT4_SECTION_4K_BIGALLOC_16K_DB_PAGE_SIZE
        int
        output yaml
        depends on SYSBENCH_EXT4_SECTION_4K_BIGALLOC_16K
        default "16384"

config SYSBENCH_EXT4_SECTION_4K_BIGALLOC_32K
        bool "ext4_4k_bigalloc_32k"
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kdevops: how do we make filesystem tests easier?

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Kconfig

LLMs today:

C coding: D-
English: B
Kconfig: A

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- Forces documentation
- Enhances determinism
- Enhances automation

Future SAT solver integration
(Boolean Satisfiability)

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kdevops: Future SAT solver integration

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Future to strive towards

```
ci-goal:
  - filesystem: xfs
  - features: [reflink, 4k]
  - testsuite: fstests
  - tests: generic/750
```

The less code or Makefiles you have to edit the better

kdevops: dynamic kconfig

SAMSUNG

- Automatically scrapes if releases file is old:
  - linux tag
  - linux-next tags
  - linux-stable tags

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    - Supports parsing your PCI tree to enable PCI passthrough

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    - Plots: A+
    - This is only getting better
    - 2024 SOTA: sysbench
    - 2025 SOTA: new world
- 2025: Port Milvus scripts
  - AI Grade in 2025: B-



Empirical evaluation of AI prompts and grading

kdevops PROMPTS.md

kdevops: Linux software 3.0 enabler

SAMSUNG

- Andrej Karpathy: AI researcher, founding member OpenAI, ex-Tesla AI lead



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- **Constrain the grammar**





Make automation a fundamental part of the architecture

kdevops: singularity for testing is here

SAMSUNG

- What is the singularity?



kdevops: singularity for testing is here

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- Borrowed from physics: you can’t see beyond the event horizon



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- I'm having a hard time prioritizing innovations
- It's not just me



kdevops: talk is cheap

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How has this been used for something productive on Linux?

Use cases

- v6.12: block size > page size for filesystems: LBS
- v6.15: block size > page size on the block layer

https://semiconductor.samsung.com/news-events/tech-blog/dividing-and-conquering-linux-high-capacity-ssd-support-with-lbs-ep1/

- TPS variability: large atomics make sense for databases
- vLLM production stack bringup is now easy



- Parallel Writeback: **212% performance improvement**

kdevops-results-archive

- We need a home to archive as much data as is possible for results
- We need infinite storage



kdevops-results-archive

- We need a home to archive as much data as is possible for results
- We need infinite storage
  - git LFS
  - rotate git trees by epoch



kdevops latest dog trick

- CI is a pain point
- Generalizing it would be amazing
- Yes you can partial vibe-code but also painful to test
- The most nasty party of automation

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- CI is a pain point
- Generalizing it would be amazing
- Yes you can partial vibe-code but also painful to test
- The most nasty party of automation
- Goal: avoid developers having to touch it
- **Generalizing CI** is possible and **done**: **Kudos to Daniel Gomez**

Turning kdevops into CI accelerator

SAMSUNG

- kdevops supports +100 profiles among different workflows
  - xfs, btrfs, ext4, tmpfs, nfs, blktests, modules, kmod, xarray, maple, firmware…

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blktests_block
blktests_loop
blktests_meta
blktests_nbd
blktests_nvme
blktests_nvmemp
blktests_scsi
blktests_srp
blktests_zbd

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btrfs_fspace
btrfs_fspace_dsync
btrfs_fspace_lzo
btrfs_fspace_zlib
btrfs_fspace_zstd
btrfs_holes
btrfs_holes_lzo
btrfs_holes_zlib
btrfs_holes_zstd
btrfs_nohofspace
btrfs_nohofspace_lzo
btrfs_nohofspace_zlib
btrfs_nohofspace_zstd
btrfs_noholes
btrfs_noholes_lzo
btrfs_noholes_zlib
btrfs_noholes_zstd
btrfs_simple
btrfs_simple_zns

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ext4_1k
ext4_2k
ext4_4k
ext4_advanced_features
ext4_bigalloc1024k_4k
ext4_bigalloc16k_4k
ext4_bigalloc2048k_4k
ext4_bigalloc32k_4k
ext4_bigalloc64k_4k
ext4_bigalloc8k_4k

ext4_defaults

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nfs-fstests-pnfs

nfs-fstests-v41

nfs-fstests-v42

nfs-gitr-pnfs

nfs-gitr-v42

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selftests-firmware

selftests-kmod

selftests-maple

selftests-module

selftests-vma

selftests-xarray

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tmpfs_default
tmpfs_huge_advise
tmpfs_huge_always
tmpfs_huge_within_size
tmpfs_noswap_huge_advise
tmpfs_noswap_huge_always
tmpfs_noswap_huge_never
tmpfs_noswap_huge_within_size

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xfs_crc
xfs_crc_logdev
xfs_crc_logdev_rtdev
xfs_crc_rtdev
xfs_crc_rtdev_extsize_28k
xfs_crc_rtdev_extsize_64k
xfs_nocrc
xfs_nocrc_16k_4ks
xfs_nocrc_1k
xfs_nocrc_2k
xfs_nocrc_32_4ks
xfs_nocrc_4k
xfs_nocrc_512
xfs_nocrc_64_4ks
xfs_nocrc_8k_4ks
xfs_reflink_1024
xfs_reflink_16k_4ks
xfs_reflink_2k
xfs_reflink_32k_4ks
xfs_reflink_4k
xfs_reflink_4k_8ks
xfs_reflink_64k_4ks
xfs_reflink_8k_4ks
xfs_reflink_dir_bsize_8k
xfs_reflink_logdev
xfs_reflink_normapbt
xfs_reflink_nrext64
xfs_reflink_stripe_len

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sysbench_postgresql_ext4_4k
sysbench_postgresql_ext4_4k_bigalloc_16k
sysbench_postgresql_ext4_4k_bigalloc_32k
sysbench_postgresql_ext4_4k_bigalloc_64k

sysbench_postgresql_xfs_reflink_16k
sysbench_postgresql_xfs_reflink_32k
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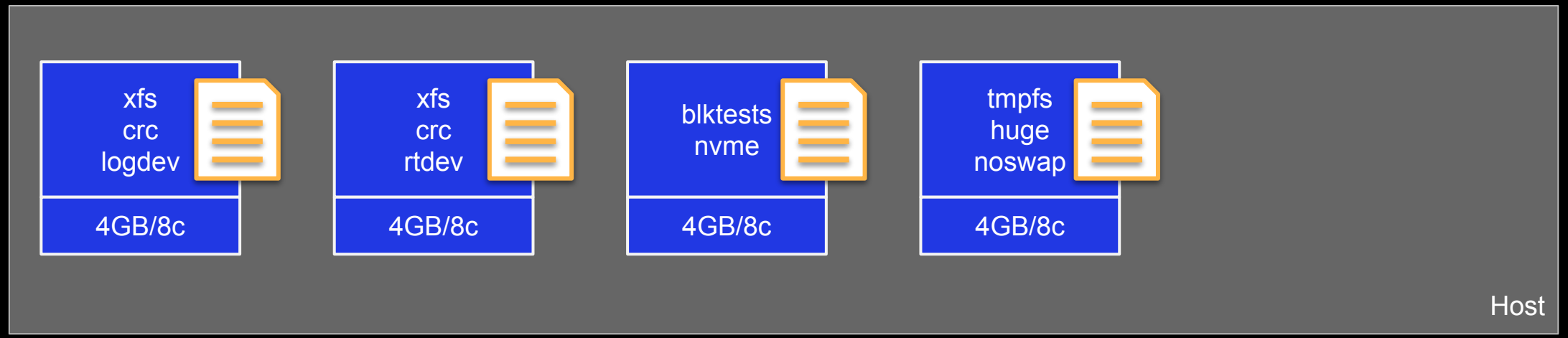
sysbench_mysql_ext4_4k
sysbench_mysql_ext4_4k_bigalloc_16k
sysbench_mysql_ext4_4k_bigalloc_32k
sysbench_mysql_ext4_4k_bigalloc_64k

sysbench_mysql_xfs_reflink_16k
sysbench_mysql_xfs_reflink_32k
sysbench_mysql_xfs_reflink_4k
sysbench_mysql_xfs_reflink_64k
sysbench_mysql_xfs_reflink_8k

Turning kdevops into CI accelerator

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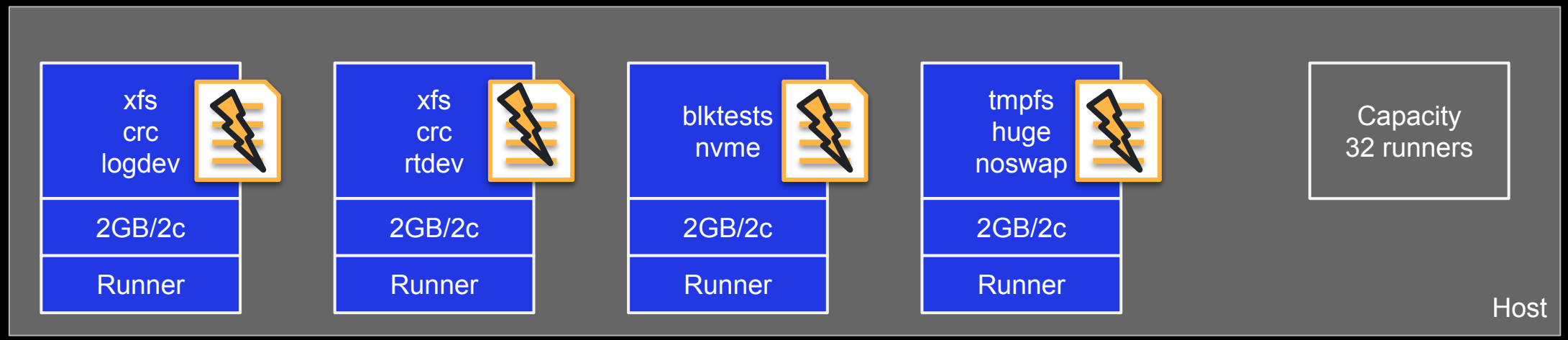
- How do we automate testing in the background? → CI
  - GitHub Actions (Reusable)
    - Easy
    - Extremely painful

Turning kdevops into CI accelerator

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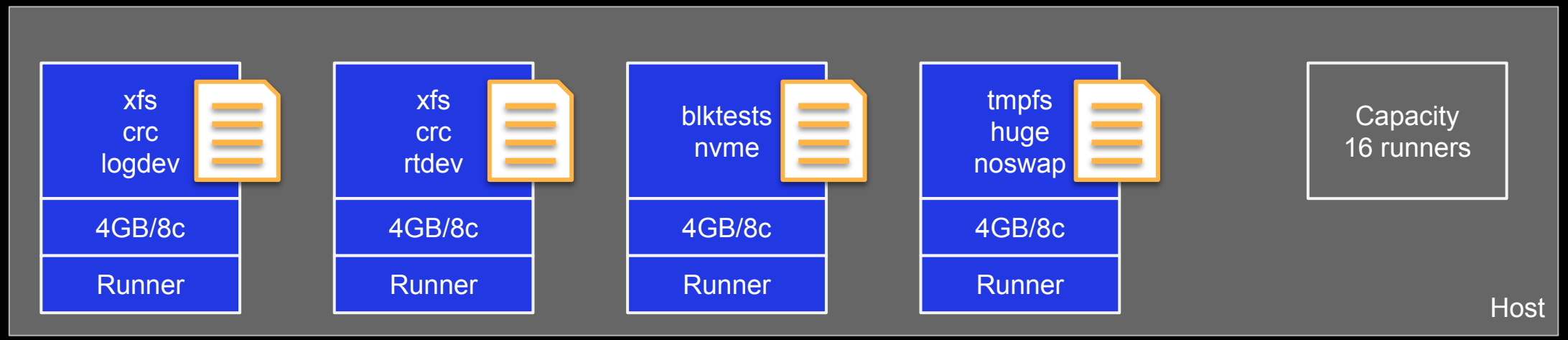
- How do we automate testing in the background? → CI
  - kdevops-ci → Test and validate kdevops framework



Turning kdevops into CI accelerator

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- kdevops supports +100 profiles among different workflows
  - xfs, btrfs, ext4, tmpfs, nfs, blktests, modules, kmod, xarray, maple, firmware…
- How do we automate testing in the background? → CI
  - kdevops-ci
  - linux-ci → Test and validate a specific profile on a specific Linux version

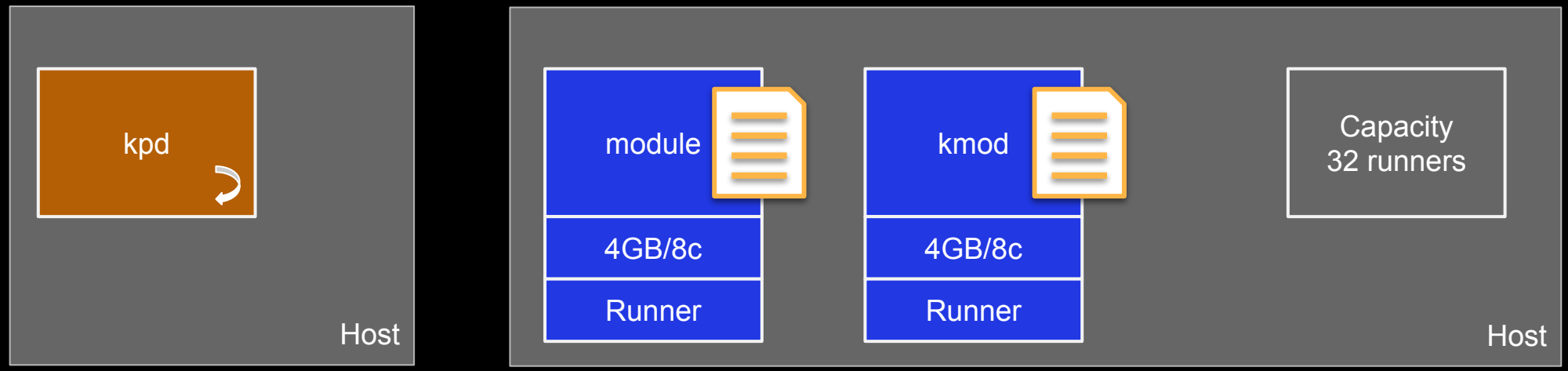


Turning kdevops into CI accelerator

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- kdevops supports +100 profiles among different workflows
  - xfs, btrfs, ext4, tmpfs, nfs, blktests, modules, kmod, xarray, maple, firmware…

- How do we automate testing in the background? → CI
  - kdevops-ci, linux-ci
  - *kpd → Test and validate a specific profile on a specific Linux version with a patch posted on lore
    - Uses kernel-patch-daemon to trigger a specific linux-ci

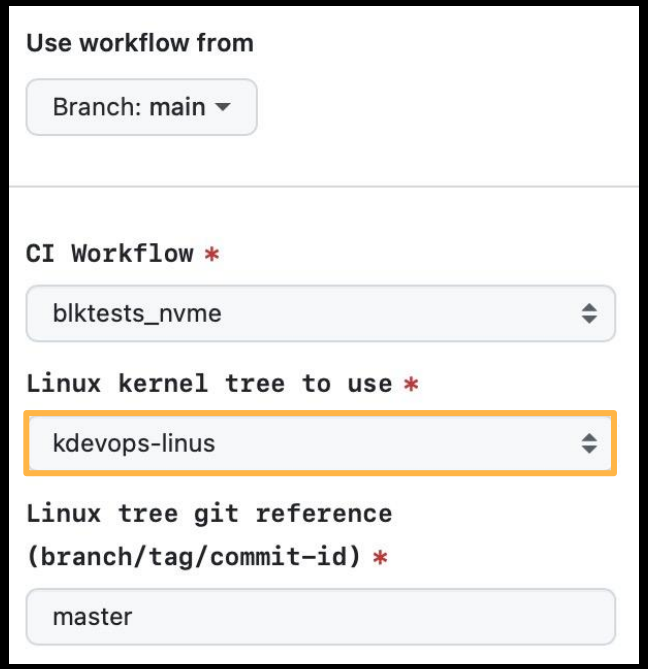


Turning kdevops into CI accelerator

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- kdevops supports +100 profiles among different workflows
  - xfs, btrfs, ext4, tmpfs, nfs, blktests, modules, kmod, xarray, maple, firmware…

- How do we automate testing in the background? → CI
  - kdevops-ci, linux-ci, *kpd
  - linux-kdevops → Test any branch with any profile using linux-ci





Turning kdevops into CI accelerator

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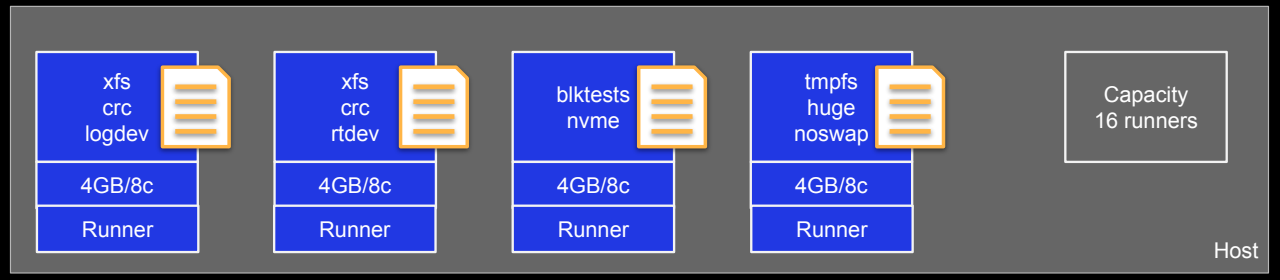
- kdevops-ci
  - Verifies every MR
  - Can be run on demand with ci-testing/** ref namespace

- linux-ci
  - Daily run of profiles against latest linux-next tag
  - Weekly run of profiles against latest Linus release or release candidate

Turning kdevops into CI accelerator

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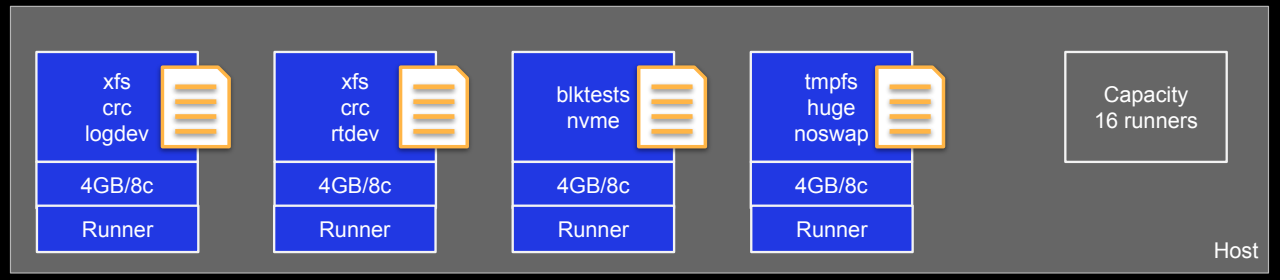
- How do we scale?
- linux-ci
  - Daily run of profiles against latest linux-next tag
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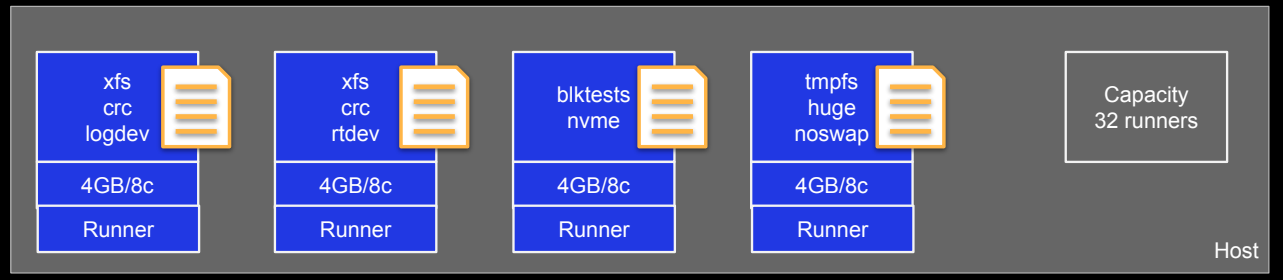


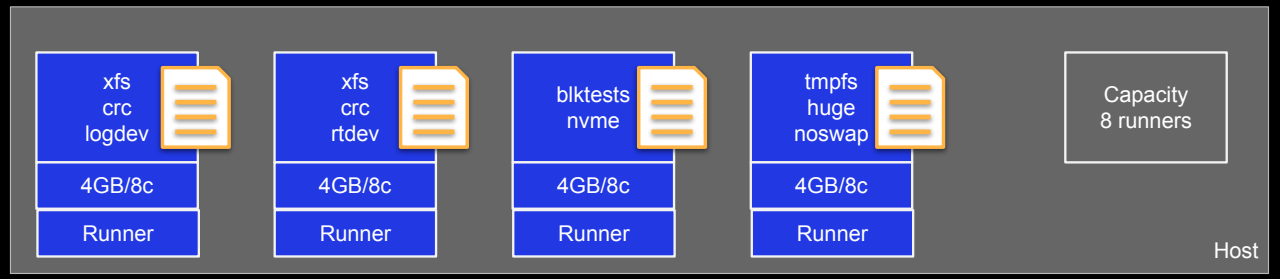
Turning kdevops into CI accelerator

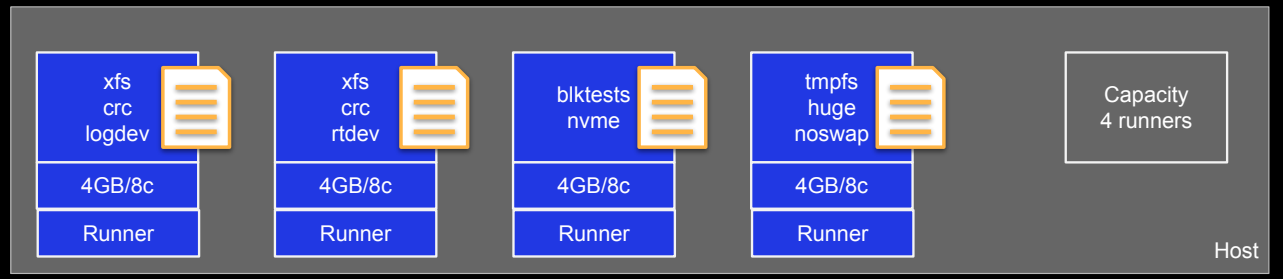
SAMSUNG

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Turning kdevops into CI accelerator

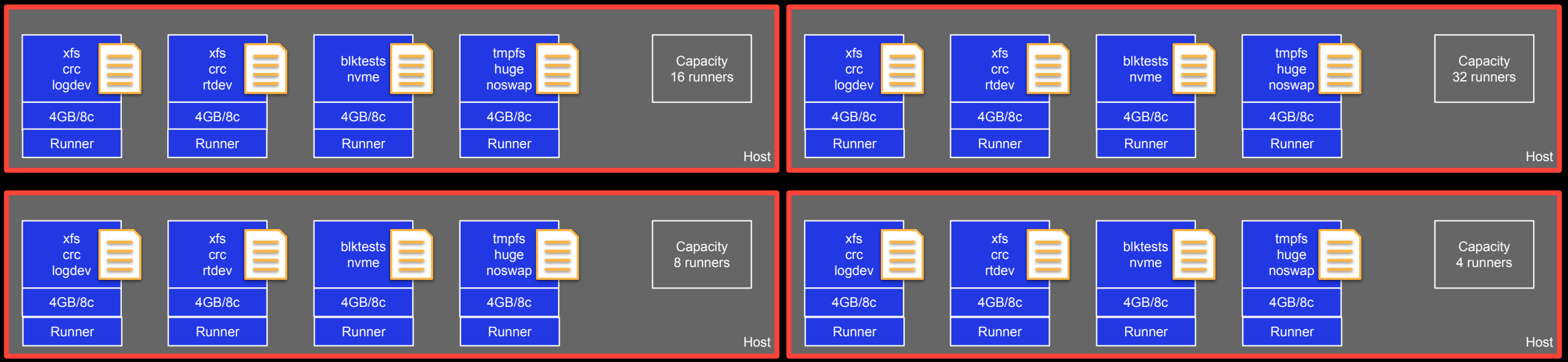
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- How do we scale?
- linux-ci
  - Daily run of profiles against latest linux-next tag
  - Weekly run of profiles against latest Linus release or release candidate
- kdevops normal workflow
  - Setup configuration (Kconfig)
  - Idempotent host setup
  - Idempotent bringup (VM creation and basic initialization)
  - Idempotent workflow installation (tooling and testing framework installation)
  - Idempotent workflow setup (testing framework configuration)
  - Linux build
  - Linux install

Turning kdevops into CI accelerator

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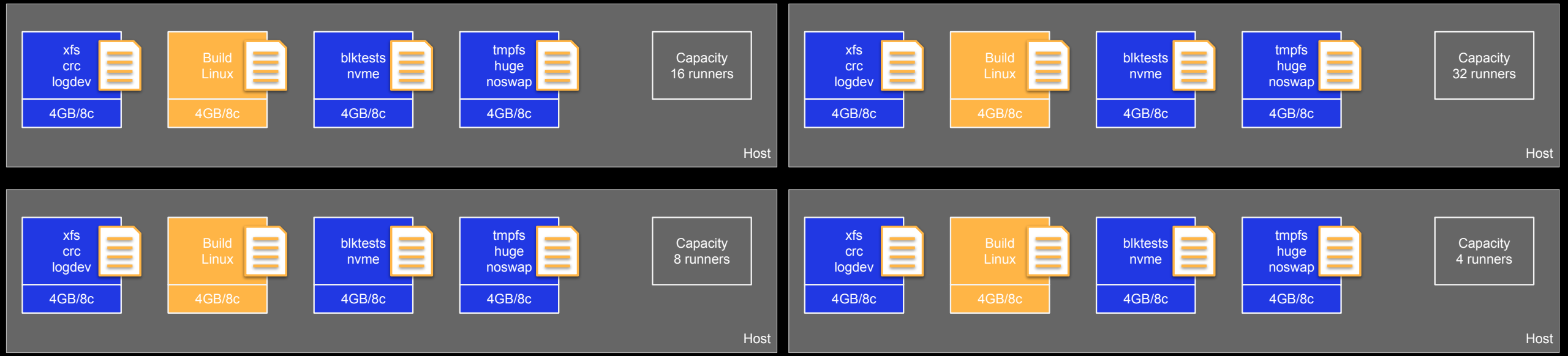
- How do we scale?
- kdevops normal workflow
  - Linux build (uses host nproc)
  - Every runner competes for the same resources



Turning kdevops into CI accelerator

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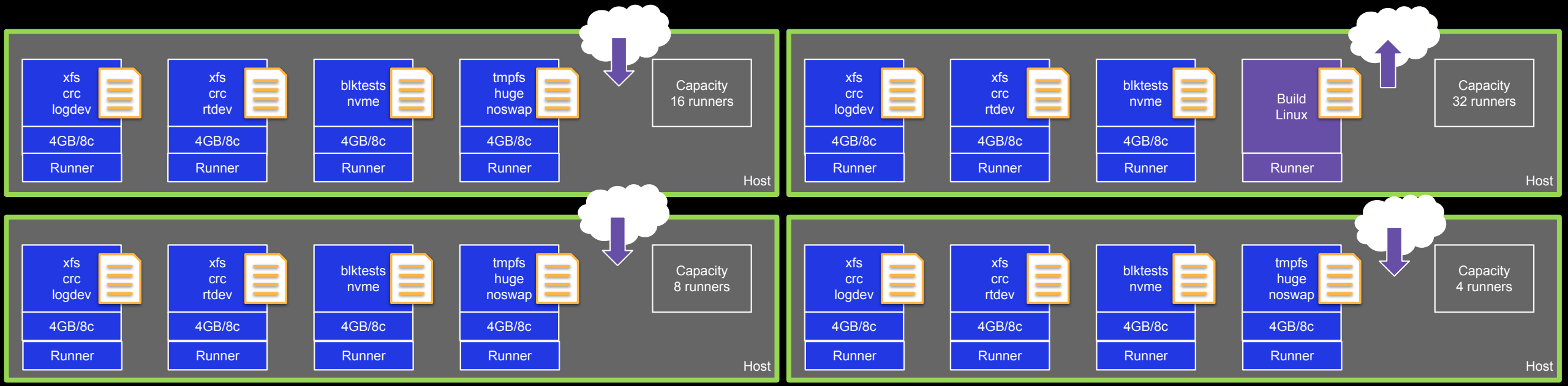
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Turning kdevops into CI accelerator

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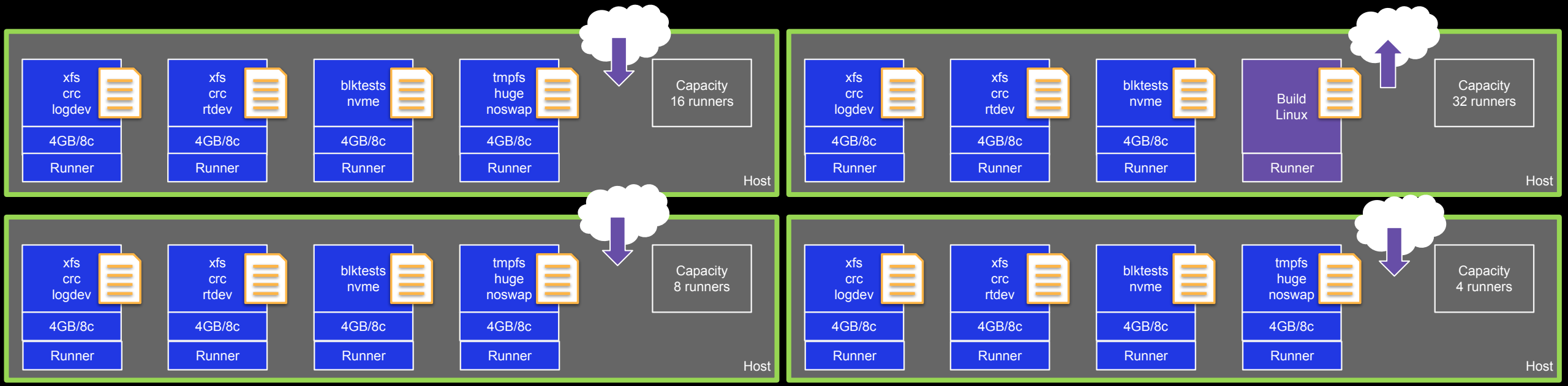
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Turning kdevops into CI accelerator

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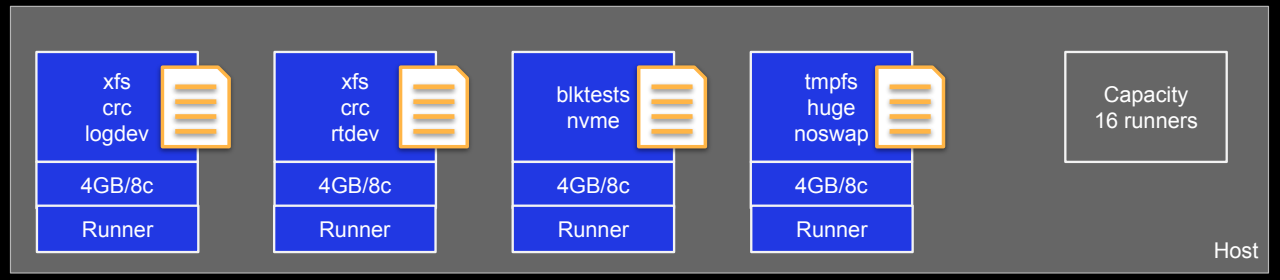
- How do we scale?
- kdevops normal workflow
  - Linux builder runner (uses host nproc)
  - Problem: No support for heterogeneous runners, scheduling preference



Turning kdevops into CI accelerator

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- How do we scale?
- linux-ci
  - Daily run of profiles against latest linux-next tag
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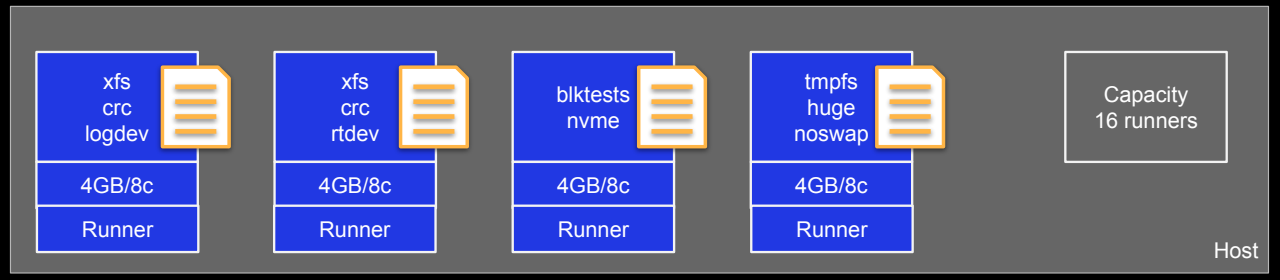


- blktests

Turning kdevops into CI accelerator

SAMSUNG

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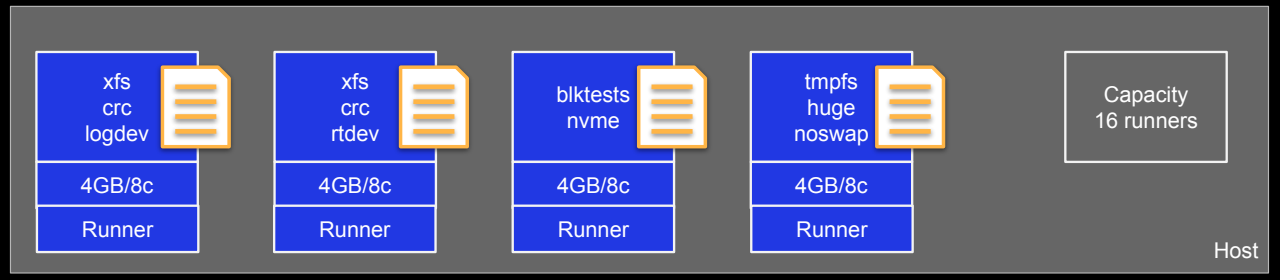


- blktests
- modules

Turning kdevops into CI accelerator

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- How do we scale?
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  - Daily run of profiles against latest linux-next tag
  - Weekly run of profiles against latest Linus release or release candidate



- blktests
- modules
- Report to mailing lists?